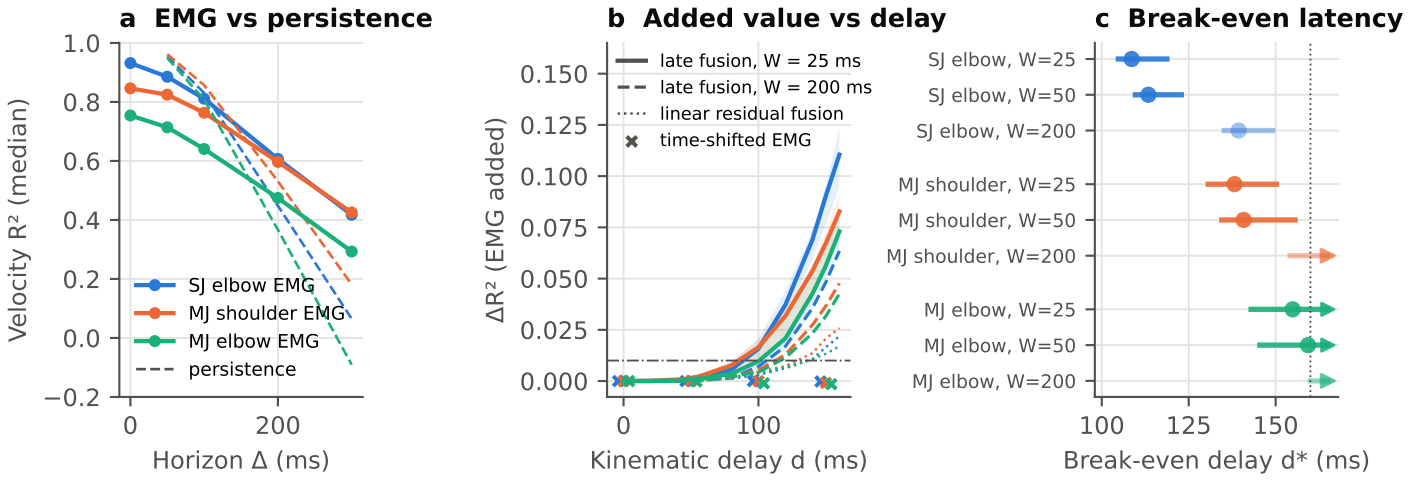
a EMG vs persistence
Velocity R² (median)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0
200
Horizon Δ (ms)
SJ elbow EMG
MJ shoulder EMG
MJ elbow EMG
persistence
b Added value vs delay
ΔR² (EMG added)
0.150
0.125
0.100
0.075
0.050
0.025
0.000
late fusion, W = 25 ms
late fusion, W = 200 ms
linear residual fusion
time-shifted EMG
0
100
Kinematic delay d (ms)
c Break-even latency
SJ elbow, W=25
SJ elbow, W=50
SJ elbow, W=200
MJ shoulder, W=25
MJ shoulder, W=50
MJ shoulder, W=200
MJ elbow, W=25
MJ elbow, W=50
MJ elbow, W=200
100
125
150
Break-even delay d* (ms)